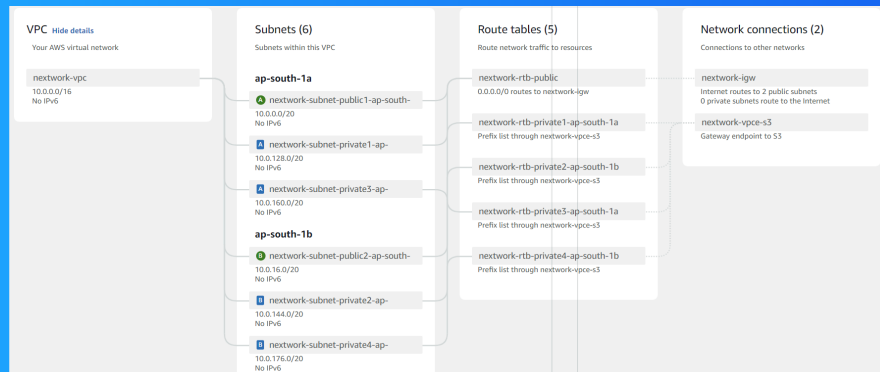




# Launching VPC Resources



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# Introducing Today's Project!

## What is Amazon VPC?

Amazon VPC is a service that lets users create isolated virtual networks in the AWS cloud. It is useful for controlling IP address ranges, subnets, and security settings, enabling secure and scalable application deployments and resource management.

## How I used Amazon VPC in this project

I used Amazon VPC in today's project to create a secure network environment for my application. Configured subnets for public and private resources and established security groups to control access and ensure data PROTECTION.

## One thing I didn't expect in this project was...

One thing I didn't expect in this project was the complexity of configuring security group rules. Balancing access between public and private subnets while ensuring proper security measures required more careful planning and testing.

## This project took me...

1 HOUR



# Setting Up Direct VM Access

Directly accessing a virtual machine means connecting to your EC2 instance via SSH (Linux) or RDP (Windows) without using intermediary services. This gives you direct control over the instance for management, configuration, and troubleshooting tasks.

## SSH is a key method for directly accessing a VM

SSH traffic means secure data communication between a client and a remote server using the Secure Shell (SSH) protocol. It ensures encrypted, authenticated connections, commonly used for securely managing remote servers like EC2 instances via command

## To enable direct access, I set up key pairs

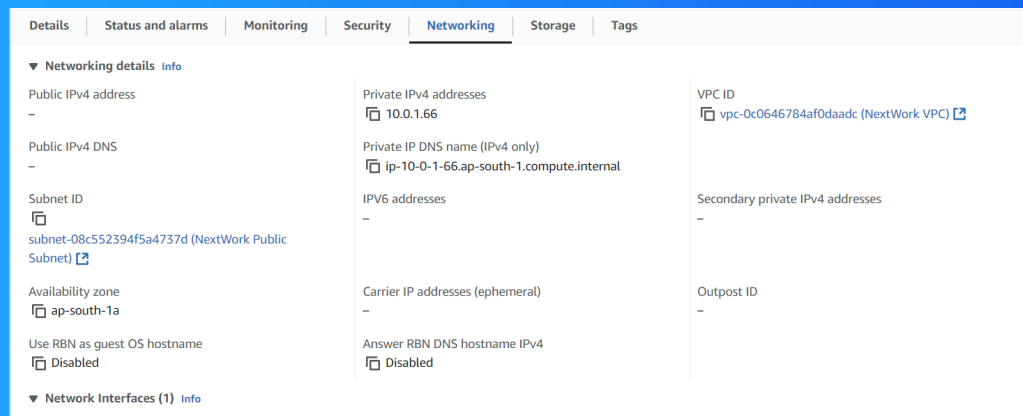
Key pairs are a set of security credentials used in AWS to securely connect to EC2 instances. They consist of a public key stored by AWS and a private key kept by the user. The private key is used to authenticate and establish secure SSH connections.

A private key's file format means the structure used to store the key. Common formats include PEM (.pem) and PPK (.ppk for PuTTY). My private key's file format was .pem, which is widely used for SSH connections to EC2 instances.



# Launching a public server

I had to change my EC2 instance's networking settings by selecting the correct VPC, subnet, and security group.



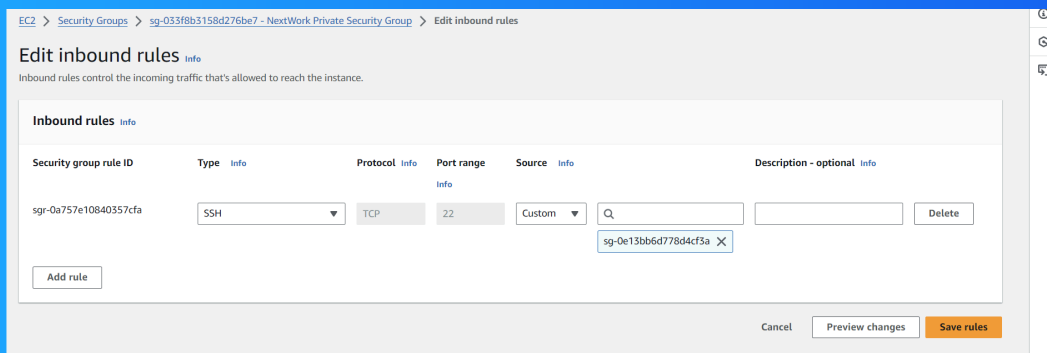




# Launching a private server

My private server has its own dedicated security group because I only want traffic from my public server to enter into private server, not from anywhere(0.0.0.0/0) on the internet.

My private server's security group's source is the security group associated with my public server, which means only traffic from the public server is allowed to access the private server, enhancing security by restricting outside access.



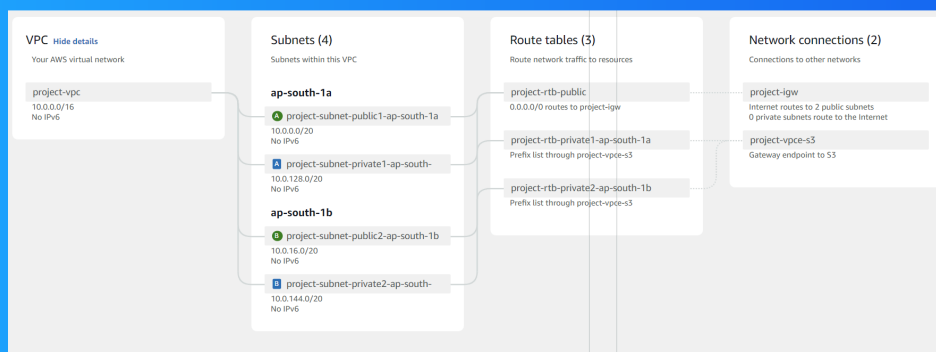


# Speeding up VPC creation

I used an alternative way to set up an Amazon VPC! This time, I utilized the "VPC and more" option to create the VPC, define subnets, set up route tables, configure an internet gateway, and establish security groups tailored to my needs.

A VPC resource map is a visual representation of the components within a VPC in AWS. It shows resources like subnets, route tables, internet gateways, and security groups, helping users understand their network architecture and connectivity.

My new VPC has a CIDR block of 10.0.0.0/24. It is possible for my new VPC to have the same IPv4 CIDR block as my existing VPC because they are in different AWS accounts or regions, allowing CIDR block overlap without causing conflicts in routing.



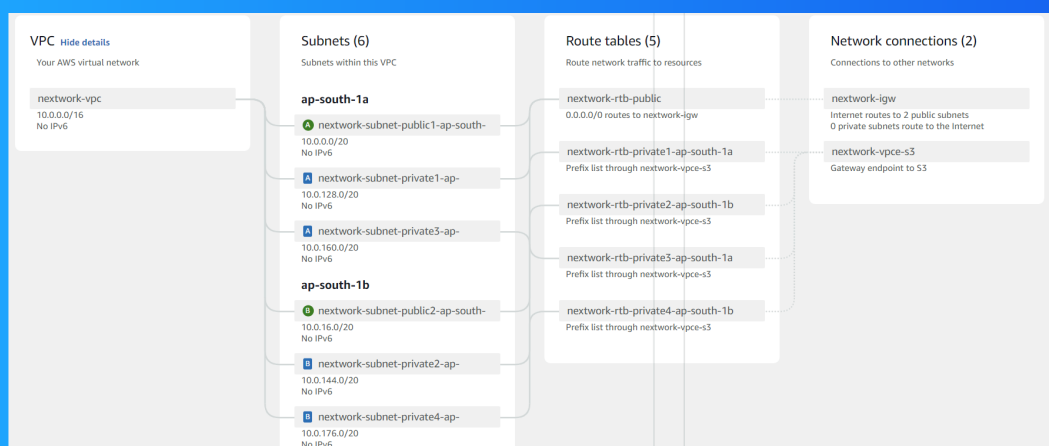


# Speeding up VPC creation

## Tips for using the VPC resource map

When determining the number of public subnets in my VPC, I only had two options: one public subnet per availability zone or more, depending on my design. This was because each public subnet needs an associated route table and internet gateway.

The setup page also offered to create NAT gateways, which are managed AWS resources that enable instances in a private subnet to access the internet for updates and external services while preventing inbound traffic from internet, enhancing security.





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